3 Fig. 3.1 shows a simplified diagram of a solar cell.

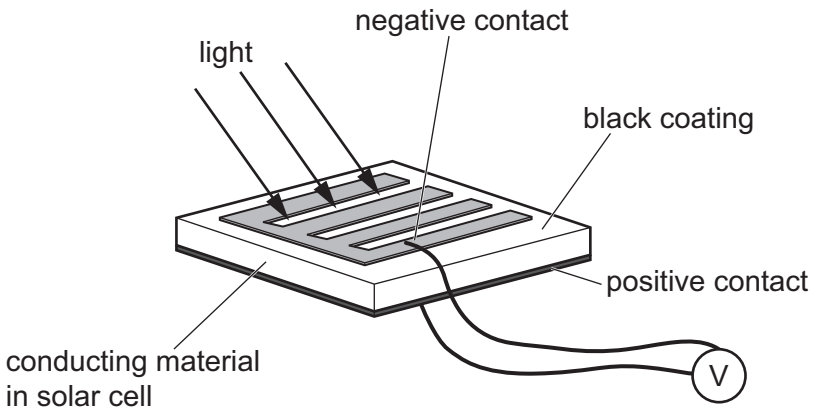

Fig. 3.1

(a) Describe the energy transfer in the solar cell.

..... [2]

(b) Suggest how the black coating allows the solar cell to transfer more energy.

.....

..... [1]

(c) 0.72 kW of light is incident on the solar cell in Fig. 3.1. The cell has an efficiency of 75%.

(i) Calculate the output power of the cell.

output power = ... [2]

(ii) State the meaning of the term kilowatt-hour (kWh).

.....

..... [1]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(iii) Energy is produced by each solar cell for an average of 6 hours per day. A household uses approximately 7400 kWh of electrical energy per year.

Calculate the number of solar cells needed to produce energy for one household. Give your answer as a whole number of solar cells.

number of solar cells = ... [3]

[Total: 9]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN